FRANCESCO GANIS

PhD, Sound & Music Engineer

francesco.ganis@gmail.com | francescoganis.github.io | Francesco Ganis | Copenhagen, DK

With a strong background in audio digital signal processing (DSP), prototyping, and the evaluation of audio solutions for people with hearing impairment, I bring experience in developing user-centered audio technologies and vibrotactile feedback systems. I am a passionate audio engineer and researcher, with deep love for music and sound. I am eager to contribute, while continuing to expand my knowledge and skills. You can find my portfolio at francescoganis.github.io.

EDUCATION

PhD, Music Training Tools for Children with Hearing Impairment

Aalborg University

Sep. 2022 – Mar. 2026 | Copenhagen, DK

Skills learned: Research, Prototyping, Evaluating

- Designing, developing, and evaluating musical training solutions for children with hearing loss, including:
  - vibrotactile feedback prototypes
  - video game with vibrotactile feedback for timbre recognition
  - scalable training tool for sound localization in children
  - training tool for emotion recognition in children

Master's Degree, Sound & Music Computing

Aalborg University

Sep. 2020 – Jun. 2022 | Copenhagen, DK

Skills learned: DSP, Prototyping, Evaluating

- Thesis: "Vibrotactile Feedback for Musical Training of Cochlear Implants Users"
- Completed courses in Sound Processing, Embodied Interaction, Music Perception and Cognition, New Interfaces for Musical Expression

Bachelor's Degree, Electronic Music & Sound Engineering

"C. Pollini" Conservatory

Sep. 2016 – Apr. 2020 | Padova, IT

Skills learned: Sound Engineering, MAX/MSP, Sound Design

- Graduated with honors
- Thesis: "Continuous Sonification of Autonomous Driving Vehicles with Purposes of Passengers Information and Trust Augmentation"

EXPERIENCE

Teaching Assistant

Aalborg University

Sep. 2022 – Aug. 2025 | Copenhagen, DK

Skills learned: Mentoring, Communication, Supervision

- Assisting students in the *Sound Processing* course
- Assisting students in the *Physical Models for Sound Synthesis* course

STRENGTHS

Technical Skills

MATLAB, Prototyping, Evaluation, DSP, Audio Measurement, C, Arduino, Git, Unity, C#, Embedded Programming, ESP32, LaTeX, Lasercutting, MAX/MSP, Pro Tools, Pure Data, Reaper, Sound Engineering, 3D Printing

Soft Skills

Analytical Thinking, Collaboration, Detail-oriented, Problem-solving, Scientific Dissemination, Supervision, Teamwork, Technical Documentation

OTHER EXPERTISE

Photography, Playing Drums

LANGUAGES

Italian ●●●●●
Mother tongue

English ●●●●●
Full professional proficiency

PUBLICATIONS

Journal Articles

- F. Ganis et al., "The role of haptics in training and games for hearing-impaired individuals: A systematic review," *Multimodal Technologies and Interaction*, vol. 8, no. 1, p. 1, 2023.

- Supervision of student projects in Medialogy and Sound & Music Computing programs

Student Helper
Aalborg University

Oct. 2021 – Apr. 2022 | Copenhagen, DK

Skills learned: Audio Programming | DSP | Web Development

- Created sound examples in MATLAB for the book *Serafin, S., Buxton, B., Gaver, B., & Bly, S. (2022). Auditory interfaces. Focal Press*
- Developed and maintained websites

Internship
Oticon Medical

Oct. 2021 – Jan. 2022 | Copenhagen, DK

Skills learned: Product Development | Prototyping | Evaluation

- Development of smartphone cover that provides high-quality vibro-tactile feedback to support music listening for individuals with hearing impairment

ICT Technician
GAMA S.r.l.

Dec. 2019 – Jan. 2020 | Tavagnacco, IT

Skills learned: Technical Support | Hardware Maintenance

- Audio and conferencing systems' installation and maintenance
- Assembling and repairing PCs, workstations and small servers

Sound Engineer
Various

2017 – 2018 | Veneto Region, IT

Skills learned: Sound Engineering | Live Electronics

- Centro D'Arte & C. Pollini Convervatory - *Live Electronics, Sound Enigneering for 11 Channels Sound System*
- RmS Recording Studio - *Boom Operator, Editing Technician*
- Bluetrain's Studio - *FoH Sound Engineer, Live Stage Technician, Live Video Director*

Internship
Danieli Automation S.p.A.

Jun. 2013 – Jul. 2013 | Buttrio, IT

Skills learned: Prototyping | Industrial Design

- Design and creation of a working prototype of a measurement system for foundry equipment

Conference Proceedings

- F. Ganis et al., "Vibrotactile teddy bear: Enhancing musical experiences for children with cochlear nerve deficiency through a vibrotactile soft toy," SMC Proceedings, 2025, pp. 115–120.
- F. Ganis et al., "Vibrotactile memory: A case study of timbre perception training in children with cochlear implants using a video game," SMC Proceedings, 2024, pp. 134–141.
- F. Ganis et al., "Tickle tuner-haptic smartphone cover for cochlear implant users' musical training," in *HAID*, Springer Nature, 2022, p. 14.
- F. Ganis et al., "Real-time timbre transfer and sound synthesis using ddsp," SMC Proceedings, 2021, pp. 175–182.

Albums

- M. Finco et al., *Six flute sonatas op.1*, album, EAN Code: 7.93588412371, Japan, 2018.

FURTHER TRAINING

Masterclass, Live Electronics
Chigiana Academy

Jul. 2019 | Siena, IT

Masterclass, Symphonic Orchestra Recording
G. A. Fano Institute

Jul. 2018 | Spilimbergo, IT

Diploma, Music, Theory and Solfège
G. Tartini Conservatory

Sep. 2009 | Trieste, IT

REFERENCES

Prof. Stefania Serafin
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Prof. Lone Percy-Smith
@ Righospitalet, Copenhagen - DK
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Prof. Nicola Bernardini
@ Conservatorio di Latina, Latina - IT
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